

Neurons - Biology Notes

Introduction to Neurons

Neurons are specialized cells of the nervous system that transmit information through electrical and chemical signals.

Structure of a Neuron

- 1 Cell Body (Soma): Contains nucleus and cytoplasm
- 2 Dendrites: Receive signals from other neurons
- 3 Axon: Conducts impulses away from the cell body
- 4 Myelin Sheath: Insulates axon and speeds up transmission
- 5 Axon Terminals: Transmit signals to other cells

Types of Neurons

- 1 Sensory Neurons: Carry signals from sensory organs to the brain
- 2 Motor Neurons: Carry signals from brain to muscles
- 3 Interneurons: Connect neurons within the brain and spinal cord

Functions of Neurons

- 1 Receive stimuli
- 2 Generate nerve impulses
- 3 Transmit signals to other neurons or muscles
- 4 Help in reflex actions and coordination

Nerve Impulse Transmission

A nerve impulse is an electrical signal that travels along the neuron.

- 1 Resting potential: Difference in charge across membrane
- 2 Action potential: Rapid change in membrane potential
- 3 Synapse: Junction between neurons where neurotransmitters are released

Synapse and Neurotransmitters

- 1 Synapse is the gap between two neurons
- 2 Neurotransmitters carry signals across synapse
- 3 Examples: Dopamine, Serotonin, Acetylcholine

Reflex Action

Reflex action is a quick automatic response to a stimulus without conscious thought.

- 1 Reflex arc includes receptor, sensory neuron, interneuron, motor neuron, and effector

Disorders Related to Neurons

- 1 Alzheimer's Disease: Memory loss due to neuron degeneration
- 2 Parkinson's Disease: Movement disorder
- 3 Multiple Sclerosis: Damage to myelin sheath

Conclusion

Neurons are essential for communication in the body. They help in thinking, movement, sensation, and coordination.